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What is a Python Library?
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=> Imagine writing a calculator.

** Without libraries: You write every mathematical function yourself.

** With libraries: Someone already wrote those functions, you simply use them.

without library

```
def square(num):
    return num*num
```

```
print(square(10))
```

Using library

```
import math
```

```
print(math.sqrt(100))
```

** Library = Ready-made code.

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Real-Life Example
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Suppose you are constructing a house.

Without Library, You manufacture

Bricks
Cement
Steel
Doors
Windows

Then build house (This is impossible because manufacturing is very difficult for me)..

** Instead of manufacturing everything, we can purchase ready-made material required for house construction.

** Programming is same, Libraries are ready-made building blocks.

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What is a Library?
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** Collection of packages

Example : Numpy Library

=> Contains

- Linear Algebra

- Random
- Statistics
- Matrix

=> package means collection of modules.

=> Module means python file.

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Types of Python Libraries
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1) Standard Libraries (Already available, we no need to install)

- math
- random
- datetime
- os
- sys
- json

2) Third Party Libraries (we need to install)

- numpy
- pandas
- matplotlib
- flask
- django
- tensorflow
- fast-api
- scikit-learn
- streamlit

Using : using pip we can install third party libraries.

Note: Better to use virtual environment to install project specific libraries.

Numpy :

- Purpose : Numerical Computing
- Used For : Arrays, Matrix Operations, Linear Algebra
- Projects: AI, ML, Scientific Computing

Pandas :

- Purpose : Data Analysis
- Used For : Reading CSV/Excel, Data Cleaning, Data Manipulation
- Projects : Data Science, Business Intelligence

Matplotlib :

- Purpose : Data Visualization
- Used For : Line, Bar, Pie, Graph, Scatter Charts
- Projects : Reports, Dashboards

Flask :

- Purpose : Web Framework
- Used For : REST APIs, Small Web applications
- Projects : Backend services, Microservices

Django :

- Purpose : Fullstack Web Framework
- Used For : Large Scale Enterprise web applications
- Projects : ECommcer, ERP, CRM

Scikit-Learn :

- Purpose : Machine Learning
- Used For : Classification, Regression, Clusttering
- Projects : Predication Systems, Recommendation Systems

TensorFlow :

- Purpose : Deep Learning
- Used For : Neural Networks, Computer Vision, NLP
- Projects : Image Recognition

FAST-API :

- Purpose : High Performance API framework
- Used For : REST APIs for AI/ML/GEN AI/Agentic AI projects
- Projects : Backend REST APIs

Streamlit :

- Purpose : Web Apps for Data Science
- Used For : Build interactive Dashboards
- Projects : AI Chatbot demos, data Visualization

Hugging Face :

- Purpose : AI Model hub
- Used For : LLMs, Text generation, image genration, AI model deployment
- Projects : ChatGPT-Like application, AI Assistants, Translation systems
